

Assignment on Unit III & IV

Submission Date: April 30, 2025

PART A: Ordinary Differential Equations

- Formulate differential equations corresponding to the following functions and hence find the order, degree, and nature (linear or non-linear) of the equation (A and B are arbitrary constants):
 - $y = Ae^{3x} + Be^{-3x}$
 - $y = e^x(A \cos x + B \sin x)$
 - $y = A \sec x + B \tan x$
- Using appropriate methodology, solve the following first order differential equations:
 - $e^y(1+x^2)\frac{dy}{dx} - 2x(1+e^y) = 0$
 - $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$
 - $(y^2 - 2xy)dx = (x^2 - 2xy)dy$
 - $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$
 - $\frac{dy}{dx} = \frac{y(x-y)}{x(2x+3y)}$
 - $\cos^2 x \frac{dy}{dx} + y = \tan x$
 - $\frac{dy}{dx} + 2xy = e^{-x^2}$
 - $\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2}$
 - $(x^2 + 1)\frac{dy}{dx} + 2xy = 4x^2$
 - $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$
 - $\frac{dy}{dx} + y \tan x = y^3 \cos x$
 - $\frac{dy}{dx} - \frac{y}{x \log x} = \frac{\log x}{x}$
 - $ydx + (-x - y^2)dy = 0$
 - $(y^2 e^{xy^2} + 4x^3)dx + (2xy e^{xy^2} - 3y^2)dy = 0$
 - $(\sin x \cos y + e^{2x})dx + (\cos x \sin y + \tan y)dy = 0$
 - $(x^3 - 3xy^2)dx + (y^3 - 3x^2y)dy = 0$, where $y(0) = 1$
 - $(3xy^2 - y^3)dx + x(xy^2 - 2x^2y)dy = 0$
- Find the value of k for which the differential equation $(xy^2 + kx^2y)dx + (x+y)x^2dy = 0$ is exact and solve it for this value of k .
- Show that the equation $3y(x^2 - 1)dx + (x^3 + 8y - 3x)dy = 0$ is exact and its particular solution when $x = 0, y = 1$ is $xy(x^2 - 3) = 4(1 - y^2)$.
- Solve the differential equation $\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$ and then find the particular solution if $x = 3$, when $t = 0$ and $x = 8$ when $t = \log 2$.
- Find the solution for the following equations:
 - $(D^2 - 3D + 2)y = e^{2x}$
 - $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = x^2 + 3$
 - $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 3x + 1$

- d. $(D^2 - D - 2)y = \sin 2x$
- e. $\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + 2y = x^2 e^{-x}$.
7. Solve the differential equation $\frac{d^2 y}{dx^2} + 9y = \sec 3x + \tan 3x$ using method of variation of parameters.
8. Use the method of change of independent variable to solve the Cauchy-Euler differential equation
- a. $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$
- b. $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = \sin(\ln x)$

PART B: Vector Calculus

9. If a force $\vec{F}(x, y) = 2x^2 y \hat{i} + 3xy \hat{j}$, displaces a particle in the xy -plane from $(0,0)$ to $(1,3)$ along a curve $y = 3x^2$. Find the work done.
10. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for the vector function $\vec{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$ along the path C given by $x = 2t, y = t, z = t^3$ from $t = 0$ to $t = 1$.
11. If $\vec{F} = y\hat{i} - x\hat{j}$, find $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,2)$ along the following path consisting of the straight lines from $(0,0)$ to $(1,0)$ and then to $(1,2)$.
12. Evaluate $\iint_S \vec{A} \cdot \hat{n} ds$, where $\vec{A} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$, and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.
13. Evaluate $\iint_S \vec{A} \cdot \hat{n} ds$, where $\vec{A} = 18xz\hat{i} - 12\hat{j} + 3y\hat{k}$, and S is the part of the plane $2x + 3y + 6z = 12$ including in the first octant.
14. Evaluate $\iiint_V \vec{F} dv$ where $\vec{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$ and V is the region bounded by the surfaces $x = 0, x = 2, y = 0, y = 4, z = x^2, z = 2$.
15. Use Green's theorem to evaluate $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the closed curve of the region bounded by $y = \sqrt{x}$ and $y = x^2$.
16. Verify Green's theorem for $\int_C (x^2 y dx + x^2 dy)$, where C is the closed curve described counter-clockwise direction of the triangle with the vertices $(0,0), (1,0), (1,1)$.
17. Use the Gauss's Divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.